

Benchmarking different multi-omic models for breast cancer subtypes classification

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Abstract—We compare three recent multi-omics models: MOFA, IntegrAO, and P-NET with a classical Baseline for classification of breast cancer subtypes, using data from 658 TCGA BRCA patients across mRNA, DNA methylation, and proteomics. Overall, none of the advanced models clearly outperformed the baseline. The baseline achieved a macro F1 score of 0.84 ± 0.06 , while IntegrAO reached 0.80 ± 0.04 and P-NET 0.84 ± 0.05 . MOFA showed substantially lower predictive performance, with a macro F1 score of 0.70 ± 0.10 . All methods struggled to accurately classify the Luminal B subtype, for which the highest mean F1 score was 0.65 ± 0.04 . MOFA provided the strongest interpretability but lower predictive performance. Our results suggest that newer biologically informed models offer improved interpretability, however they do not yet provide substantial gains in subtype prediction accuracy over simpler approaches.

I. INTRODUCTION

Breast cancers are among the leading causes of cancer-related death in women [1]. Treatment outcomes vary significantly by tumour subtype, making accurate subtype identification critical for informed treatment decisions. Modern cancer datasets often include multiple *omics*, which are different types of molecular measurements collected from the same tumour, such as gene expression, DNA methylation, or protein levels. Different omics layers capture complementary biological patterns, with a single omic layer often being insufficient to fully characterize complex diseases, motivating the adoption of multi-omic analyses. Classical approaches combine unsupervised dimensionality reduction applied separately to each omic layer, such as PCA, with supervised models like logistic regression or SVM. More recently, new and more complex approaches have emerged, including deep neural networks that mimic biological structures or unsupervised PCA-like methods. Benchmarking is crucial to objectively compare classical and advanced methods, assessing their performance and interpretability. The goal of our project is to assess three recent models: MOFA [2], IntegrAO [3] and P-NET [4].

II. DATA

The four models were trained and tested on a dataset of 658 unique patients extracted from a larger database, the Cancer Genome Atlas-Breast Invasive Carcinoma (TCGA-BRCA) [5], which contains breast cancer tissue samples from a cohort of over 1,000 patients, classified into four tumour subtypes: Luminal A (LumA), Luminal B (LumB), Basal and HER2-positive (HER2). Our dataset contains three types of features that measure the biological effects of breast cancer (BRCA) across the following three omic layers:

- Transcriptomics (mRNA): gene expression/activity levels.

- Epigenetics (DNAm): DNA methylation levels at different genomic sites.
- Proteomics (RPPA): protein abundance levels.

Data had been previously processed in complimentary bioinformatic pipelines, but we performed additional post-processing steps to ensure comparable scales and homoscedasticity. An initial exploratory analysis done over each omic layer reveals their distribution (see fig. 1). They exhibit distinct distributions, with mRNA showing a bimodal distribution, while DNAm and RPPA display symmetric profiles, and DNAm alone having a relatively flat distribution.

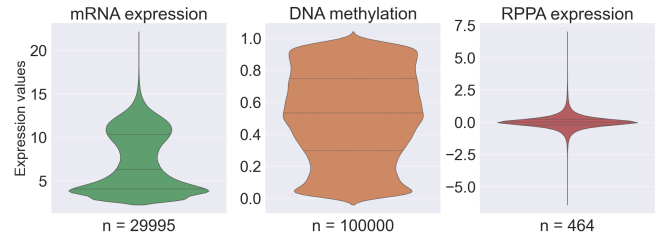


Fig. 1. Violin plots per omic layer. On the x-axis, n is the corresponding number of features. Notice the feature imbalance.

Additionally, we notice standard challenges in medical datasets: in our subset of patients, more than one third of them are missing data from at least one omic layer, and there is a considerable imbalance in both patients (fig. 2) and features (see fig. 1), which may diminish the performance of the models on under-represented subtypes. Furthermore, Uniform Manifold Approximation and Projection[6] (UMAP) visualizations fail to cluster separately the LumA, LumB and HER2 (see App. A), so we infer that less sophisticated models may struggle to properly differentiate them.

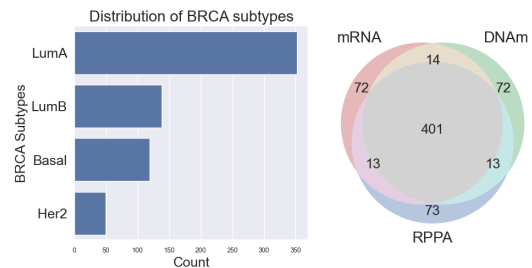


Fig. 2. **Left:** Number of patients per BRCA subtype, exhibiting class imbalance **Right:** Venn diagram showing the overlap amongst patients with complete mRNA (red), DNAm (green), and RPPA (blue) profiles.

III. MODELS AND METHODS

We now present and describe the specific methodology for each model. For the models that cannot handle missing values on their own (baseline and P-NET), the following imputation strategies were tested: removing samples with missing data, median imputation and k -nearest neighbours (k -NN) imputation. To avoid cross-omic data leakage, these imputation methods are applied on each omic layer individually instead of on the whole dataset.

A. Baseline

We developed a standard supervised learning pipeline. After dealing with the missing data, we applied PCA and standardisation, before training the model with multinomial logistic regression, using a balanced version of the SAGA [7] solver. For PCA, the number of principal components retained 70% of the total cumulative variance. After identifying the imputation method with the best performance–complexity trade-off, we maximised its performance by refining the threshold, while keeping runtime below that of the other models.

B. MOFA

MOFA is an unsupervised method that generalises PCA to multi-omics data by decomposing multiple matrices into a shared latent factor matrix Z and modality-specific weight matrices W . It was fitted on the training set using default parameters as in [2][8], and naturally ignores missing values. Since MOFA does not provide automatic projection for new samples [4], test-set latent factors were inferred manually. For each sample i , we solved the least-squares problem $Y_i = WZ_i$, where Y_i contains the non-missing features of sample i , and W is the corresponding subset of the trained weight matrix. To avoid bias, the training-set factors were recomputed using the same projection procedure rather than using the factors produced during MOFA fitting. These projected factors were then used to train and evaluate the supervised model, consistent with the baseline. Factors showing an absolute average (over all fold) correlation lower than 0.1 with all one-hot vectors of tumour subtypes were filtered out before training the supervised model.

C. IntegrAO

IntegrAO [3] is a multi-omics graph neural network designed to combine heterogeneous biological datasets. The model takes aligned multi-omics feature matrices as input and integrates them through a hierarchical network structure to predict sample level outcomes, such as cancer subtypes. It first performs an unsupervised training phase and then fine tunes using the labeled data. Beyond prediction, IntegrAO provides interpretable feature importance scores. It can handle multi-omics data even if not all patients have every type of omic. However, for each individual omic, all samples must have complete data. We used the IntegrAO package, available on the Python package Index. We used the model on a fixed selection of the most variable features for each omic, as suggested in the original paper and the standard hyperparameters given in

the official tutorial [9]. In the context of this model, no more processing of the data was needed.

D. P-NET

P-NET is an artificial feedforward neural network with a biologically informed architecture, where each node encodes a biological entity. The input layer represents features, the second layer encodes a set of genes of interest (provided as a list of genes from the P-NET paper [10]), and higher layers represent a hierarchy of pathways and biological processes. Connections between layers are constrained to follow known child–parent relationships among features, genes, and pathways.[4] The model is implemented in PyTorch with ReLU activation using the library and functions provided by Ryan et al. [11].[10]

To compare these four models, we used 5-fold cross-validation, stratified over tumour subtypes. For each fold (except in baseline), the training set was limited to the 2000 features with highest variance for each omic layer, due to limitations in computational resources. Model performance was evaluated using accuracy, macro F1 score across all classes, as well as interpretability, which we define as the ability of a model to identify the factors driving the prediction process and to explain the relationship between inputs and outputs. When applicable, interpretability was assessed on the fold that achieved the best macro F1 score.

IV. RESULTS

A. Performance

We summarise in fig. 3 the macro F1 scores obtained for each model, where we dropped missing values as imputation strategy whenever this was necessary, as this turned out to have the best mean F1 score.

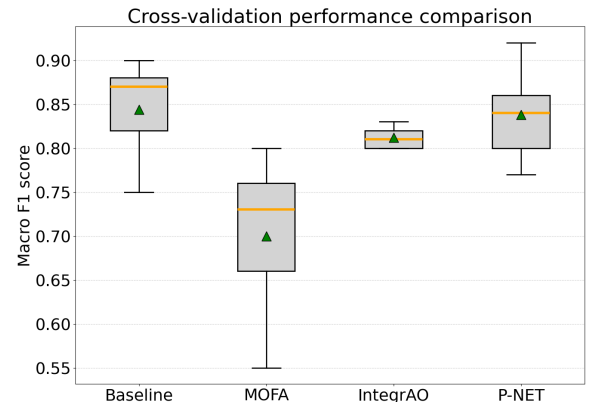


Fig. 3. Overview of the macro F1 scores for all models

fig. 4 shows the more precise distribution of per-fold F1 scores for the same models.

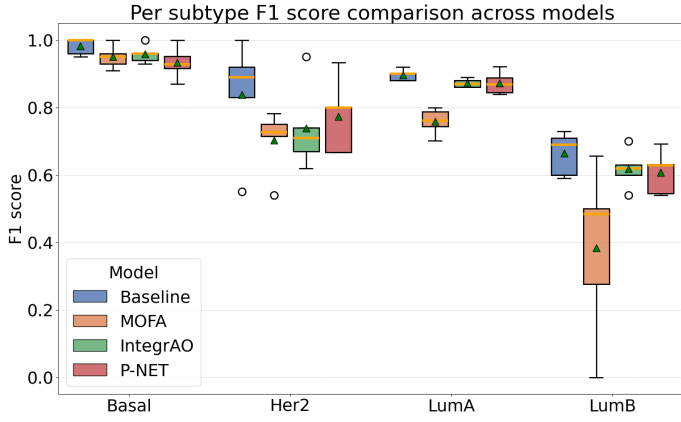


Fig. 4. Overview of the per Subtype F1 scores for all models

While performance is consistently high for the Basal and Luminal A subtypes, all models exhibit much lower and more variable performance on the Luminal B subtype. Note that the baseline shows high variability on the Her2 subtype with a lowest score of 0.55. MOFA’s performance is also quite low and variable on the LumB subtype.

For the baseline, we note that dropping missing values performed well due to the considerable number of patients without missing data. Median and nearest neighbours imputation were more consistent, but had worse average performance and higher computational cost (see fig. 5). Thus, we choose the former as a reference for comparison. Tuning the threshold of explained variance, we obtained a mean macro F1 score of 0.84 ± 0.06 , and a mean accuracy of 0.86 ± 0.03 .

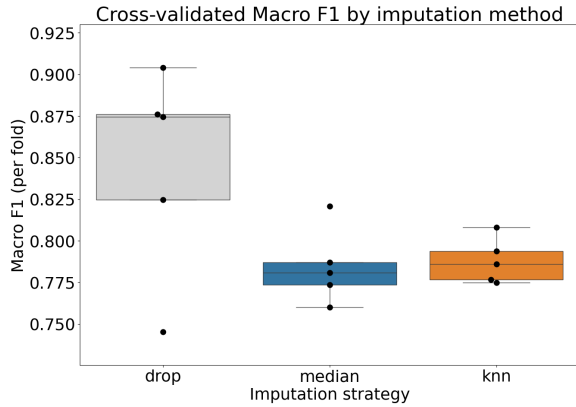


Fig. 5. Boxplots of macro f1-scores per imputation method.

Compared to the baseline, MOFA performed poorly. With a Macro F1 score of 0.70 ± 0.10 and an accuracy of 0.72 ± 0.06 , it may look like MOFA is not a useful model. However, it has a strong interpretability that is discussed later.

For IntegrAO, one advantage is that it can still make predictions when only one or two omics are available for a sample. However, the best performance of a macro F1 score of 0.80 ± 0.04 for IntegrAO was achieved by inferring on all available omics. The model’s importance scores for each feature can be found in the Appendix D.

Finally, for P-NET, the results obtained for the different imputations methods are given in Table I.

Imputation method	Accuracy	Macro F1 score
Drop NaNs	0.82 ± 0.02	0.85 ± 0.03
Median Imp	0.81 ± 0.03	0.83 ± 0.03
k -NN Imp.	0.80 ± 0.02	0.82 ± 0.02

TABLE I
F1-SCORE COMPARISON USING DIFFERENT TYPES OF IMPUTATION FOR P-NET.

Similarly to the baseline, the various implementations of P-NET that we evaluated did not yield substantial improvements in predictive performance. Across 5-fold cross-validation, when dropping missing values, P-NET achieved a macro F1 score of 0.84 ± 0.05 , which is comparable to the baseline and slightly higher than IntegrAO.

Overall, these results show that even if these models do not clearly perform better than a simple baseline for subtype classification, their ability to explain which biological features drive the predictions still makes them useful for multi-omics analysis. The detailed model outputs of importance of the features can be found in Appendix E.

B. Interpretability

MOFA produced a set of 14 uncorrelated factors, among which we could identify those that show significant and specific associations with tumour subtypes. Generally, these factors are able to differentiate the tumour subtypes from each other, although the LumA and LumB clusters remain largely intermixed (Appendix C).

For each factor, MOFA provides feature loadings, allowing us to identify the most contributive features from each omics layer (fig. 6, A,B,C). By combining information about feature loadings with the association of factors to tumour subtypes (e.g., factor logistic regression class weights, Pearson correlation) (fig. 6,D) we can reconstruct the relationship between omics features and tumour subtypes and identify subtype-specific omic patterns.

In brief, a feature strongly associated with a factor will also be strongly associated with the tumour subtypes linked to that factor. The sign of this association can be considered as the product of the signs of the feature-factor and factor-subtype associations.

The Baseline, similar to MOFA, allows the reconstruction of these relationships as it provides feature loadings for each principal component (PC), allowing the identification of the most contributive features for a given PC. It also provides logistic regression class weights for the PCs, enabling the identification of components that are strongly associated with specific tumour subtypes. (Appendix B)

Both P-NET and IntegrAO facilitate the identification of input features that drive tumour subtype predictions and rank them by average importance in order to identify the most contributive one from each omic layer separately (IntegrAO)

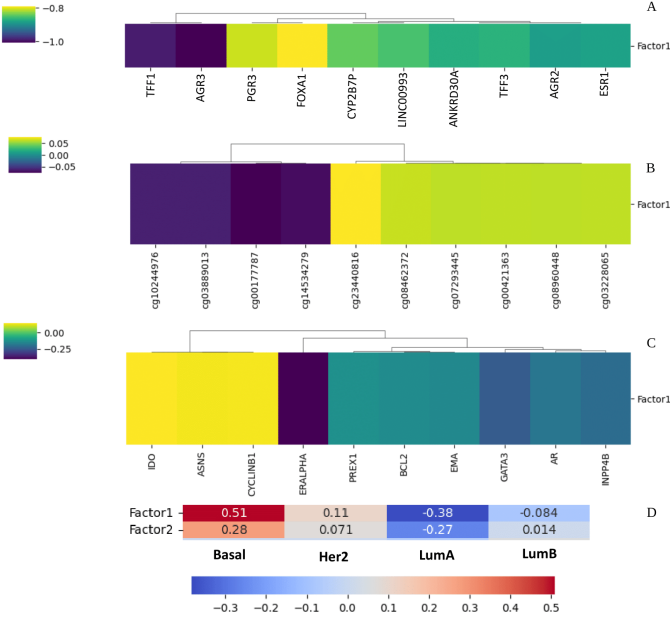


Fig. 6. Overview of MOFA interpretability: Loadings of top 10 most contributive feature for Factor 1 from mRNA(A), DNAm(B),RPPA(C) and First two rows of the Factor-tumour subtype correlations matrix .

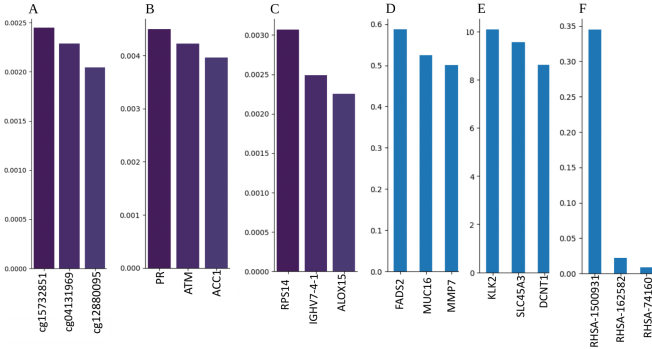


Fig. 7. Overview of Neural Networks interpretability: Top 3 most contributive feature from omic layer from DNAm(A), RPPA(B),mRNA(C) ranked by IntegrAO by Absolute mean importance and top 3 most important input features across all layer (D), most important genes(E), and pathways from last hidden layer(F) ranked by P-NET according to average importance

(fig. 7, A,B,C) or from all layers considered together (P-NET) (fig. 7D). In addition, P-NET has the capacity to represent the feature space in terms of genes (first hidden layer) and pathways (remaining hidden layers) (fig. 7,E,F) which are also ranked by average importance. This helps to identify those features that are important for discriminating the different tumour subtypes. However, contrary to MOFA and the baseline method, they do not provide information about the positive and negative association relationships that could exist between specific features and specific tumour subtypes.

Among the tested integration models, none were able to surpass the baseline in terms of performance. Specifically, P-NET showed comparable performance to the baseline, whereas IntegrAO and MOFA led to significantly lower results, as in fig. 3. However, these initial findings must be interpreted in light of the experimental setup: all models were trained using default parameters and on datasets of limited size. Although the dataset size per fold (approximately 500 samples with 2000 features) is sufficient for model training, the strong dimensionality reduction applied limits the ability of deep learning models to fully exploit the high-dimensional feature space, thereby reducing their potential advantage over simpler methods. Further benchmarking studies are necessary, including comprehensive parameter tuning and experiments on datasets of varied sizes and feature dimensionalities, to determine whether different performance trends may emerge.

Regarding interpretability, all models were able to identify features relevant for tumour classification. However, only MOFA and the Baseline could infer specific association relationships between features and distinct tumour subtypes. (fig.6, fig. 7, Appendix B).Moreover, MOFA has an interpretability advantage over the baseline, as its latent factors capture cross-omic variance, whereas principal components only represent within-omic variance.

Thus, in the case where most patients of the dataset have complete omics profiles, the model presenting the best performance-interpretability trade-off would be P-NET if the goal is to discover genes and pathways relevant for discriminating tumour subtype; or the Baseline, if the primary interest is to identify specific molecular patterns that are positively or negatively associated with distinct tumour subtypes.

Notice that using imputation for patients with missing omic layers generally decreases model performance compared to simply dropping those patient(fig.5, tab. I). Thus, in the case of a dataset where many patients lack one or more omic layer, MOFA and IntegrAO are preferable as they can ignore missing omics layers during training. This avoid the need for imputation, and thereby maximizes patient retention. In this scenario, IntegrAO is preferable if predictive performance is the priority, while MOFA is more suitable if the interpretability is the primary interest.

Regarding implementation, P-NET was by far the easiest to set up thanks to [11], MOFA was more complex but well documented, and integrAO was the most difficult to implement because of the lack of documentation. However, using the implementations [11] and from our GitHub [12], both P-NET and integrAO can be applied relatively quickly to a new dataset, whereas MOFA generally requires a full analysis from scratch each time. The training of MOFA can also be significantly longer than the other two. In our case, MOFA took around 1 hour to cross validate, while for P-NET and integrAO a few minutes were enough using a GPU.

ACKNOWLEDGEMENTS

We would like to thank Fellay Lab for hosting the project, and in special Dr. Barry John Ryan and Mariam Ait Oumelloul for their qualitative guidance and advice through the project.

CODE AVAILABILITY

A GitHub was developed for this project, available at <https://github.com/CS-433/project-2-multiomicmethods>.

VI. ETHICAL RISKS

The primary ethical risk in our study arises from the demographic imbalance of the underlying dataset in terms of sex and ethnicity, as highlighted in fig. 8.

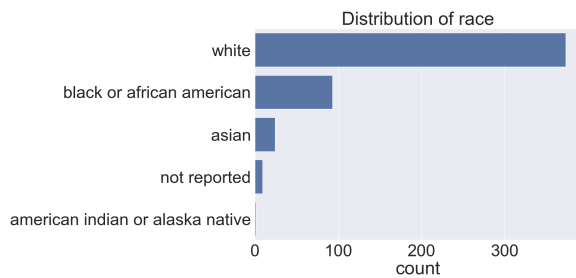


Fig. 8. Bar plot of ethnicity of patients

The absence of male cases could be a limitation, but its ethical significance is relatively modest: male breast cancer represents less than 1% of all diagnoses, and exhibits notably distinct characteristics from its female counterpart [13]. In particular, it would be advisable to carry out a dedicated study for male BRCA rather than obtain proportional sampling.

By contrast, the under-representation of racial and ethnic minorities is more consequential. Even if omic data is not dependent on ethnicity in general, factors such as ancestry-associated variation could introduce bias in the trained models if minority subgroups are insufficiently represented in the samples [14].

Given that our study focuses on the evaluation of models, we do not attempt to correct these demographic skews within the existing datasets. In addition, the options we have considered (increasing our dataset or removing over-represented samples) are either too complex to implement or too restrictive, severely limiting the precision of our results.

However, it is essential for downstream users to account for these imbalances when applying the models: one should assess subgroup performance and, if needed, implement strategies such as targeted sampling. More broadly, efforts to improve the diversity of genomic datasets are critical for ensuring that our models perform in a equitable, generalizable, and clinically reliable way across populations.

REFERENCES

- [1] "World cancer report: Cancer research for cancer prevention," 2020, licence: CC BY-NC-ND 3.0 IGO. [Online]. Available: <https://publications.iarc.who.int/Non-Series-Publications/World-Cancer-Reports/World-Cancer-Report-Cancer-Research-For-Cancer-Prevention-2020>

- [2] R. Argelaguet, B. Velten, D. Arnol, S. Dietrich, T. Zenz, J. C. Marioni, F. Buettner, W. Huber, and O. Stegle, "Multi-omics factor analysis—a framework for unsupervised integration of multi-omics data sets," *Molecular systems biology*, vol. 14, no. 6, p. e8124, 2018.
- [3] S. Ma, A. G. Zeng, B. Haibe-Kains, A. Goldenberg, J. E. Dick, and B. Wang, "Moving towards genome-wide data integration for patient stratification with integrate any omics," *Nature Machine Intelligence*, vol. 7, no. 1, pp. 29–42, 2025.
- [4] H. A. Elmarakeby, J. Hwang, R. Arafeh, J. Crowdis, S. Gang, D. Liu, S. H. AlDubayan, K. Salari, S. Kregel, C. Richter *et al.*, "Biologically informed deep neural network for prostate cancer discovery," *Nature*, vol. 598, no. 7880, pp. 348–352, 2021.
- [5] W. Lingle, B. J. Erickson, M. L. Zuley, R. Jarosz, E. Bonaccio, J. Filippini, J. M. Net, L. Levi, E. A. Morris, G. G. Figler, P. Elnajjar, S. Kirk, Y. Lee, M. Giger, and N. Gruszauskas, "The cancer genome atlas breast invasive carcinoma collection (tcga-brca)," 2016, data set. [Online]. Available: <https://doi.org/10.7937/K9/TCIA.2016.AB2NAZRP>
- [6] L. McInnes, J. Healy, and J. Melville, "Umap: Uniform manifold approximation and projection for dimension reduction," 2020. [Online]. Available: <https://arxiv.org/abs/1802.03426>
- [7] A. Defazio, F. Bach, and S. Lacoste-Julien, "Saga: A fast incremental gradient method with support for non-strongly convex composite objectives," 2014. [Online]. Available: <https://arxiv.org/abs/1407.0202>
- [8] Ricard Argelaguet, Britta Velten, Damien Arnol, Sascha Dietrich, Thorsten Zenz, John C Marioni, Florian Buettner, Wolfgang Huber Oliver Stegle . (2018) Multi-omics factor analysis—a framework for unsupervised integration of multi-omics data sets. Accessed: 16.12.2025. [Online]. Available: <https://link.springer.com/article/10.15252/msb.20178124>
- [9] S. Ma, A. G. Zeng, B. Haibe-Kains, A. Goldenberg, J. E. Dick, and B. Wang, "Integrao," <https://github.com/bowang-lab/IntegrAO>, 2024, accessed: 08.12.2025.
- [10] Chaeun Lee, Barry Ryan, Riccardo E. Marioni, Pasquale Minervini, Ian Simpson . (2025) Combining clinical embeddings with multi-omic features for improved patient classification and interpretability in parkinson's disease. Accessed: 16.12.2025. [Online]. Available: <https://www.medrxiv.org/content/10.1101/2025.01.17.25320664v1>
- [11] B. Ryan, "Pnet torch," <https://barry8197.github.io/PNetBook/intro.html>, 2023, accessed: 05.12.2025.
- [12] A. Z. Igor Jomaron, Fabien Donnet-Monay, "project-2-multiomicmethods," <https://github.com/CS-433/project-2-multiomicmethods>, 2025.
- [13] A. Gucalp, T. Traina, J. Eisner, and et al., "Male breast cancer: a disease distinct from female breast cancer," *Breast Cancer Research and Treatment*, vol. 173, no. 1, pp. 37–48, 2019.
- [14] J. Roelands, R. Mall, H. Almeer, and et al., "Ancestry-associated transcriptomic profiles of breast cancer in patients of african, arab, and european ancestry," *npj Breast Cancer*, vol. 7, p. 10, 2021.

Our appendix consists mainly of additional plots we made to visualise and study our data. While they may make

APPENDIX B
BASELINE

APPENDIX A
DATA EXPLORATION

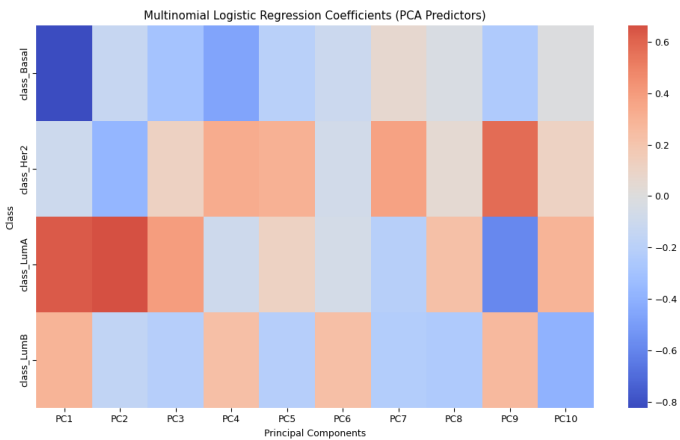
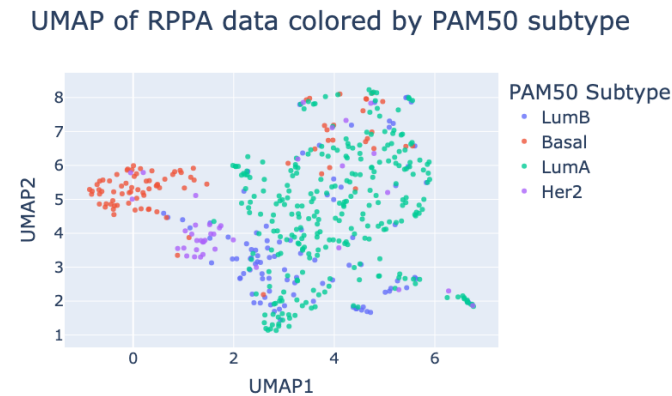
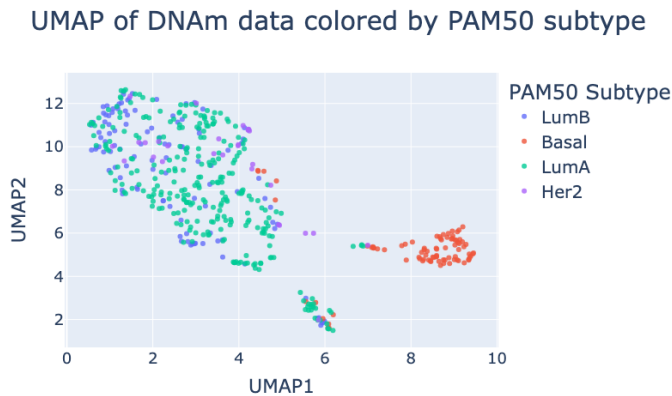
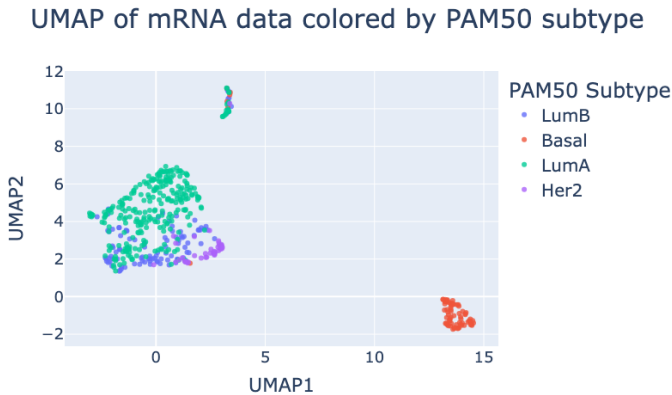
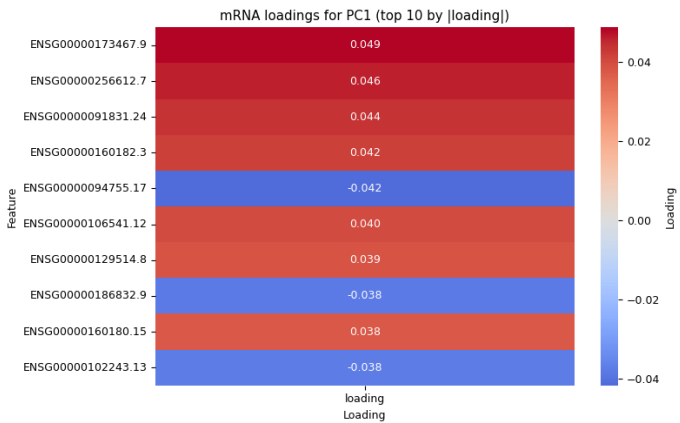


Fig. 9. Weights from logistic regression for 10 first mRNA principal component



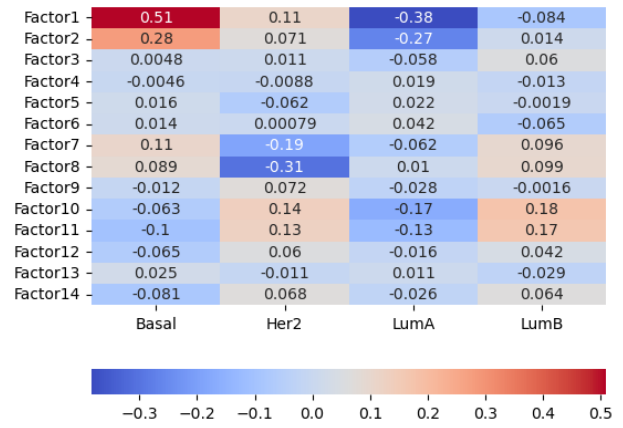
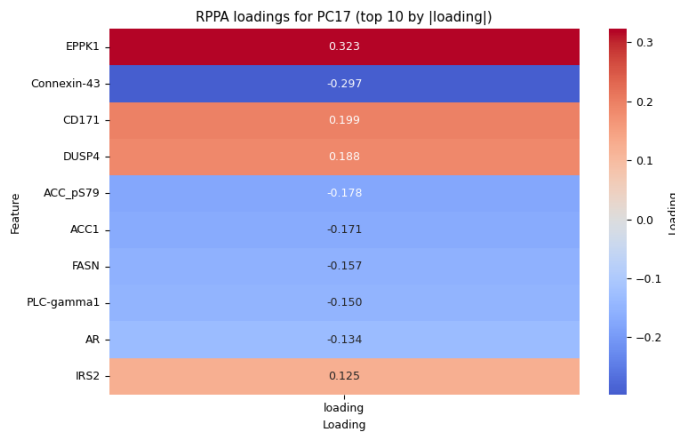


Fig. 11. Average factor-subtype correlations over the 5 folds

APPENDIX C MOFA

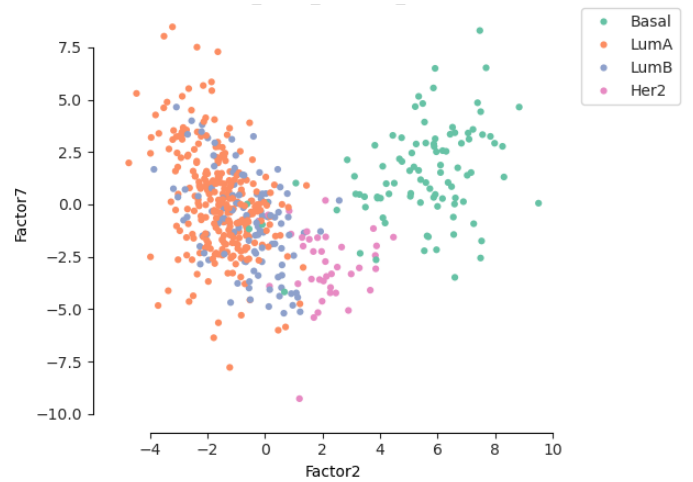


Fig. 12. Visualisation of sample clustering according to Factor 2 and 7

APPENDIX D INTEGRAO

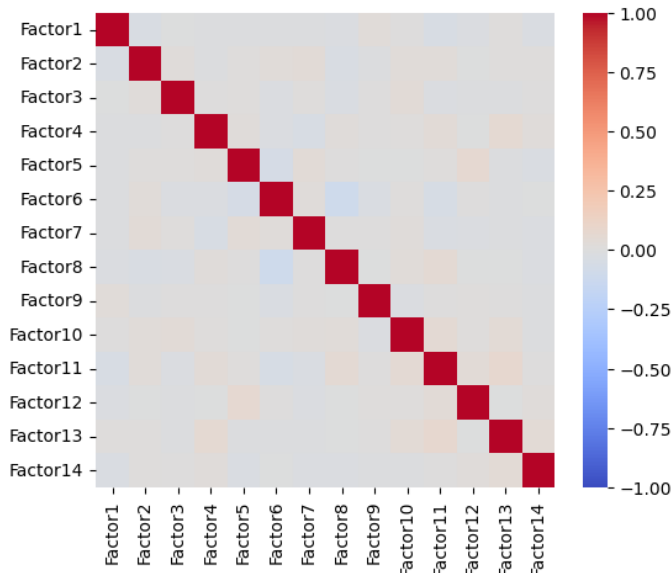


Fig. 10. Average factor-factor correlations over the 5 folds

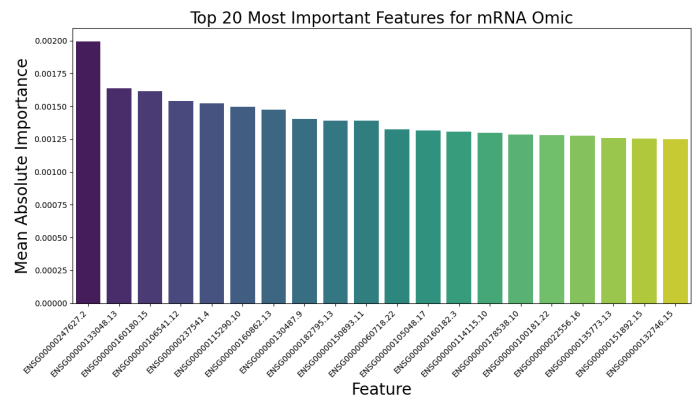
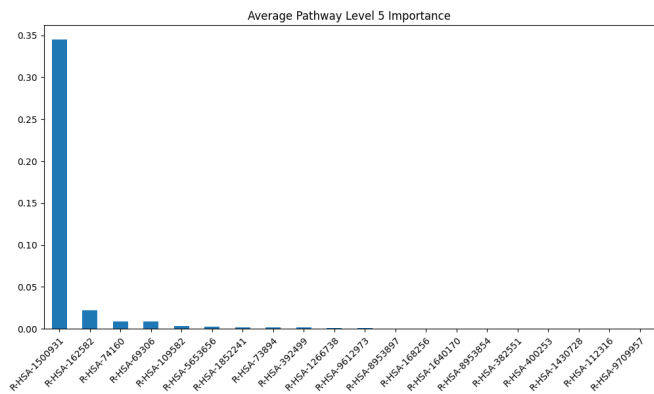
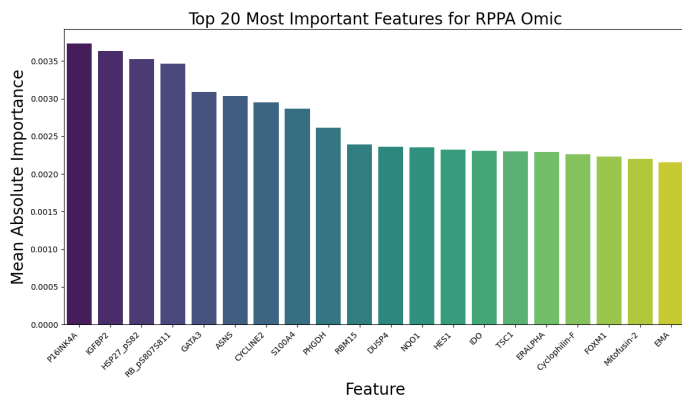
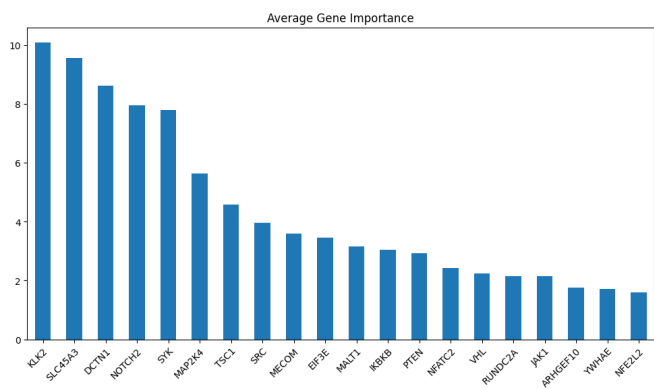
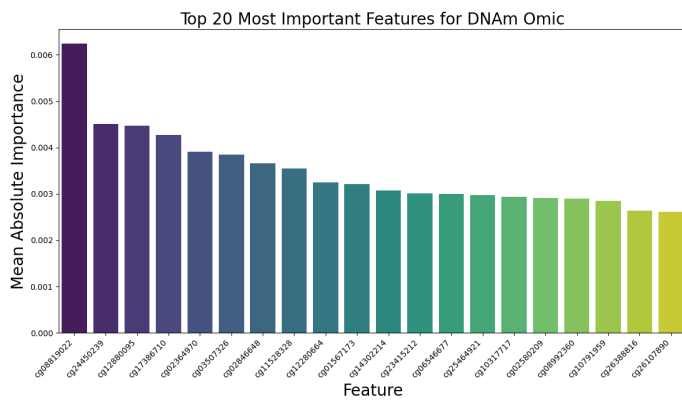


Fig. 13. Most important features given by IntegrAO for each omic.



APPENDIX E P-NET

